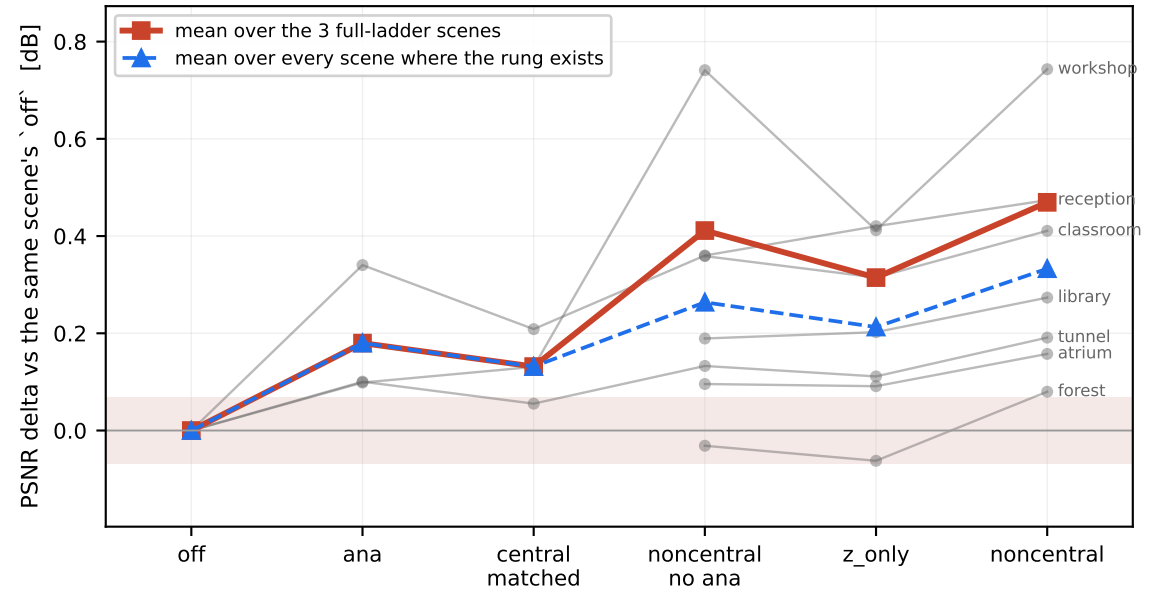
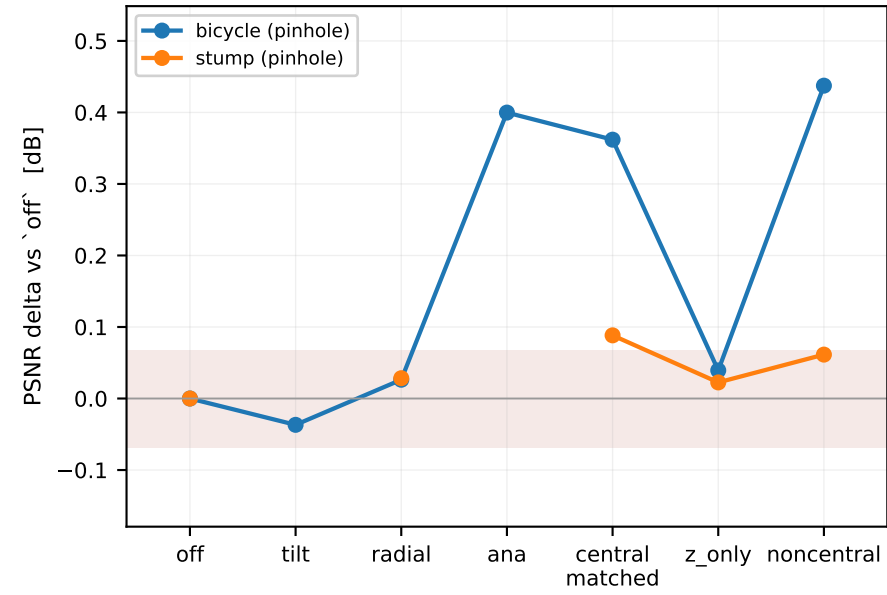


Ablation ladder: which term of the camera model carries the gain, and is the comparison interpretable? A pinhole camera cannot have a non-central pupil, so `z\_only` on pinhole is a negative control whose correct answer is zero. The two families are NEVER averaged: only deltas against each scene's own `off`.

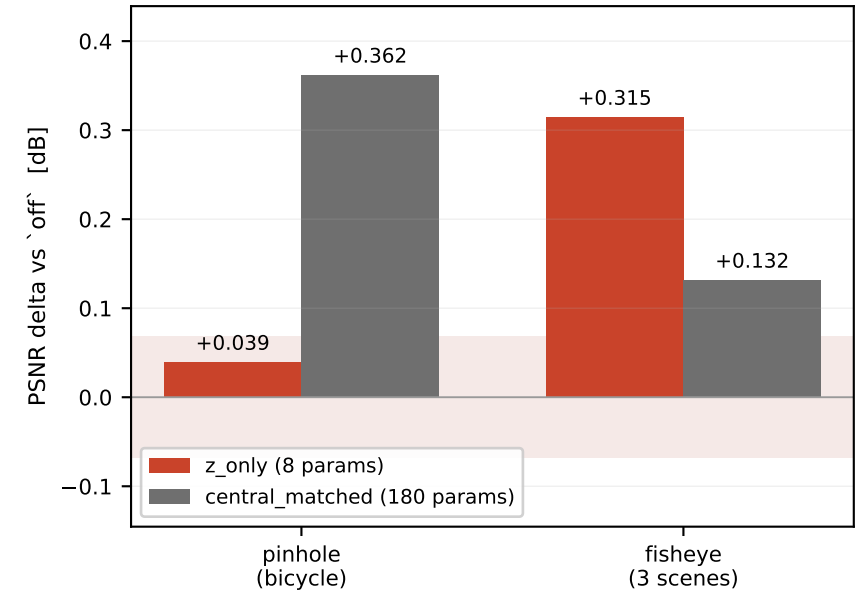
A. fisheye — myscenes, rtpf, masked $r=0.95$, $-r$ 4



B. pinhole — mip-NeRF 360, full frame



C. the 2x2 — the order INVERTS between families



shaded band = the measured 0.068 dB run-to-run noise floor; $n = 1$ run per cell, no seeds, so any $|\delta|$ inside the band is a zero.
 regularisation audit (exact penalty recomputed from the saved weights): the L2 + curvature term is 0.0000-0.139 % of the data loss over 37 runs, and `z\_only` (8 regularised params) pays 13.7x MORE than `central\_matched` (180) on workshop.
 The 'bigger rungs are penalised more' confound is therefore falsified twice over: the term is inert, and its sign is inverted.